

Muhammad Talha

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Education

University of Engineering and Technology, Lahore

Bachelor of Science in Computer Science

Sep 2023 - Present

Lahore, Pakistan

Cumulative GPA: 3.59/4

Relevant Coursework

- Artificial Intelligence
- Data Structures and Algorithms
- Object Oriented Programming
- Software Engineering
- Applied Probability and Statistics
- Database Systems
- Computer Vision
- Linear Algebra
- Multivariate Calculus

Experience

AI/ML Intern

Jul 2025 – Aug 2025

LYSA Labs (Remote)

- Implemented and fine-tuned a CNN model on the MNIST dataset, optimizing hyperparameters for improved digit recognition performance.
- Designed and developed an automated web scraping solution for the Taiwan Market Observation Post System (MOPS) to collect structured financial and disclosure data.
- Developed and deployed machine learning APIs using FastAPI for model inference and integration.
- Completed a 100 Days of Machine Learning course, gaining hands-on experience with model architectures, training workflows, and backend ML concepts.

Freelance Software Developer

Oct 2025 – Present

Fiverr (Remote)

- Delivered custom software and web solutions, including database-driven systems and responsive user interfaces.
- Developed Windows Forms and Console applications using C# and .NET.
- Built full-stack web applications using HTML, CSS, JavaScript, and the MERN stack.
- Performed machine learning model tuning and optimization on custom datasets.

Projects

SecureVision - Real-Time Face Detection & Intruder Alert System

YOLO-based real-time face detection system with AI-powered insights

- Developed a real-time face detection system using YOLOv8 with duplicate prevention via face encoding comparison.
- Built an interactive Streamlit dashboard with live camera feed, detection logs, and Excel export functionality.
- Integrated Google Gemini API to generate AI-powered facial descriptions and implemented audio alerts for intruder detection.
- Technologies: Python, YOLOv8, Streamlit, OpenCV, face_recognition, Gemini API, Pandas.

Income Prediction ML Project - Streamlit ML App

Machine learning web app for income prediction and recommendations

- Developed a Streamlit app using a Random Forest model (87% accuracy) for income classification.
- Implemented data preprocessing, model integration, and real-time predictions.
- Integrated Gemini API for personalized recommendations and improved UX with dynamic UI.
- Technologies: Python, Streamlit, Scikit-learn, Pandas, Gemini API.

Mnist Digit Classifier — Streamlit Deep Learning App

CNN-powered web app for handwritten digit recognition

- Built a Streamlit web app using a CNN trained on the MNIST dataset (70,000 images) for digit recognition.
- Implemented dual input modes: image upload and an interactive drawable canvas for real-time predictions.
- Handled full image preprocessing pipeline including grayscale conversion, resizing to 28x28, normalization, and inversion.
- Technologies: Python, TensorFlow, Streamlit, OpenCV, NumPy, Pillow.

ZenZone - Android Focus & Habit App

Kotlin Android app for focus tracking and productivity

- Built a focus-tracking app using Kotlin and MVVM (ViewModel, LiveData) with clean architecture.
- Designed responsive UI with Fragments, RecyclerView, Material Components, and custom views.
- Implemented session tracking, stats visualization, and JSON-based local storage.
- Technologies: Kotlin, Android SDK, Jetpack, Coroutines, MPAndroidChart.

Solitaire Klondike Game

C# WinForms implementation of classic card game

- Developed a fully interactive Solitaire game with complete gameplay logic.
- Implemented card movement, drag-and-drop, auto-flip, and win detection.
- Designed modular OOP-based structure for deck and pile management.
- Technologies: C#, WinForms, .NET.

Database Management System - C++ & React

Full-stack database system with custom storage and query engine

- Built a DBMS with B+ Tree indexing supporting efficient CRUD operations.
- Implemented SQL-like query parser with filtering, joins, and relational constraints.
- Developed React-based query interface with real-time execution.
- Technologies: C++, React, Node.js, B+ Trees, file-based storage.

Inventory Management System - C# (.NET, WinForms, Console)

Role-based inventory and billing management system backed by SQL

- Built a full-featured inventory system with role-based access control for Admin, Employee, and Customer roles.
- Implemented product, user, and billing management modules with a clean and intuitive WinForms UI.
- Designed the system architecture using OOP principles to ensure scalability and long-term maintainability.
- Integrated SQL for persistent data storage and efficient querying across all system modules.
- Technologies: C#, .NET, WinForms, Console, SQL, OOP.

Skills

Technical Skills

Languages/Tools: Python, C#, C++, JavaScript, TypeScript, HTML, SCSS, CSS, SQL, MongoDB

Frameworks/Libraries: Angular, React, .NET Core, ASP.NET Web API, SignalR, Tailwind CSS, PyTorch, TensorFlow, OpenCV, NumPy, Selenium, Puppeteer, Playwright

Developer/Design Tools: Visual Studio, VS Code, Postman, Git, GitHub, Google Colab, Canva, Figma, Azure DevOps, MSSQL, MongoDB Compass, n8n

Certifications and Courses

- **Microsoft Azure AI Fundamentals** – Microsoft
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- **Intro to Database & SQL** – Great Learning Academy
[View Certificate](#)